

Autonomous IT Incident Resolution Agent Using Agentic AI

Introduction: Modern enterprises operate on increasingly complex IT infrastructures spanning cloud platforms, on-premise systems, microservices, and third-party integrations. When incidents occur, the cost of downtime can reach millions of dollars per hour, yet most organizations still rely on manual, reactive processes to detect and resolve issues. Coupled with a growing global shortage of skilled IT professionals and widespread alert fatigue, traditional incident management approaches are no longer sustainable.

Objectives: This project aims to design and develop an Agentic AI-powered IT Incident Resolution system that autonomously detects, diagnoses, and resolves IT incidents in real time — reducing downtime, operational costs, and the burden on human IT teams.

Proposed Solution: The proposed solution leverages agentic AI — an advanced form of artificial intelligence capable of independent reasoning, decision-making, and executing multi-step actions. The system continuously monitors IT infrastructure across servers, networks, applications, and databases. Upon detecting an anomaly, the agent investigates root causes by analyzing logs, correlating events, and referencing historical patterns. It then autonomously executes appropriate remediation actions. For incidents requiring human judgment, the system intelligently escalates with a fully prepared diagnostic report, enabling engineers to act immediately rather than investigate from scratch.

Conclusion: By automating the end-to-end incident management lifecycle, this solution has the potential to reduce average resolution time by up to 70%, significantly lower operational costs, and improve overall IT reliability — enabling organizations to focus human expertise where it truly matters.